A Lidar Benchmark for Desert Rover Traversability: Slip Face Detection, Hazard Layers and Route Planning from Coarse Elevation Models on Active Sand Dunes

Huda Mueen
Independent Researcher
United Arab Emirates
hudamueen2407@gmail.com

Awaiz Ahmed
Dept. of Software Engineering
Al Ain University
Abu Dhabi, United Arab Emirates
awaiz42ad@gmail.com

***Abstract*—For most deserts the only free elevation models a small wheeled rover can plan on have a posting of about 30 m, while the slip faces of active dunes are often only metres long. On six North American dune fields with one metre airborne lidar as truth, we scored hazard layers from slope and interpolation to boosted trees, convolutional networks, super resolution and imagery fusion against a lidar face map, and used them in a slip aware planner at operating points chosen on the other five sites. On 10 m block means of the lidar the best layer raises mission success from 17 % to 64 %, and vehicle success (post hoc), which also requires every step to be climbable, from 9 % to 47 %. On 30 m block means no computable layer adds more than 5.3 points to 8 % (at most 2.5 points of vehicle success), and interpolation on three real 30 m products adds no more than 2.0 points. A face map oracle reaches 81 % on 30 m maps, but only 17 % vehicle success. In exploratory closed loop runs with line of sight sensing, 30 m priors raise success from 30 % to between 34 % and 48 %, and not clearly when the endpoints come from the coarse map. Code and the prediction log are public.**

***Index Terms*—field robotics, traversability, airborne lidar, digital elevation models, sand dunes, slip faces, hazard mapping, path planning, benchmark**

I. Introduction

Small wheeled ground robots are proposed for survey, inspection and search in sand deserts, including the dune seas of the Arabian Peninsula. Their routes are planned on a prior map, and for most deserts the only free digital elevation models (DEMs) are global products with a posting of about 30 m [1], [2], [3]. The dominant hazard on active dunes is the slip face, the lee side of a dune held near the angle of repose of dry sand, about 30 to 34 degrees [4], where wheel slip rises until a small vehicle can no longer climb [5], [6]. By count most face patches are shorter than one 30 m cell, although they hold a small share of the steep ground (Section IV-B), so a 30 m cell can hold a face while its mean slope is gentle (Fig. 1).

It is well established that a coarse grid understates slope [7]. It is not known how much this matters for a planner, which hazard layer to compute from a coarse map, how fine a map must be, and whether imagery, another product or onboard sensing changes the answer. We answer these questions on six North American dune fields where one metre airborne lidar provides the truth. The contributions are (1) a benchmark that scores hazard layers against a lidar face map whose meaning is the same at every resolution, on idealised surfaces from 2 to 60 m, five real elevation products and imagery; (2) a comparison of twelve families of hazard layers, learned ones trained leave one site out; (3) a planning evaluation at held out operating points with two kinds of endpoints, scored by mission success and, post hoc, by vehicle success; and (4) an exploratory closed loop with line of sight sensing, and an analysis of detection against face length and grid spacing. Table I lists the notation.

II. Related Work

Overhead priors. Airborne lidar and imagery at metre scale have served as route priors for ground vehicles [8]; drones can map ahead of the robot [9] or learn terrain properties from the air [10], and planetary missions plan from orbital data [11].

Slip on sand. Geometric costmaps score slope and roughness on a grid [12], [13]. On loose sand slip rises steeply with slope [5], [14], [15]; slip can be learned from appearance [16], desert vehicles have planned on overhead and onboard maps [17], and planners propagate slip uncertainty [18], assuming that the map resolves the slope.

Dunes in elevation models. Dune morphometry from DEMs and imagery is well developed [19], [20], and super resolution can sharpen coarse DEMs [21]. Global 30 m products have been ranked against lidar [22]; TanDEM-X, the radar behind GLO-30, is weaker over sand [23], AW3D30 comes from optical stereo [2], and DEM error in a dune sea grows above 15 degrees [24]. We know of no benchmark that scores coarse map hazard layers and planners for slip faces against lidar truth across resolutions.

III. Benchmark

A. Sites, truth and label

Six windows of one metre bare earth lidar from the 3D Elevation Program (3DEP) of the U.S. Geological Survey (USGS) [25] cover active dune fields (Table II); the Dumont

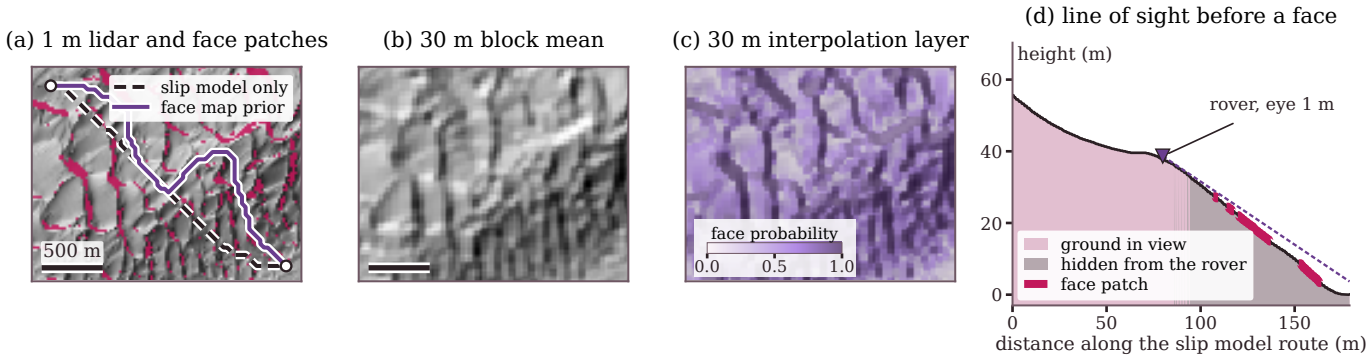


Fig. 1. Slip faces lost on a 30 m map, at Great Sand Dunes. (a) One metre lidar hillshade with face patches (rose) and the 30 m routes of one E1 pair: the slip model alone (dashed) crosses faces that a face map prior (solid) avoids. (b) The 30 m block mean of the same ground, the map the planner sees. (c) The calibrated interpolation layer computed from it. North is up. (d) Lidar profile along the slip model route (vertical scale exaggerated): from an eye 1 m above the ground, 40 m before the rover reaches a face, the face lies in ground hidden behind the convex brink ahead.

TABLE I
NOTATION (FULL GLOSSARY: SUPPLEMENT TABLE S3).

I10, I30	lidar block means at 10 and 30 m (idealised inputs)
R10, N30, G30	USGS 1/3 arc second DEM, NASADEM, Copernicus GLO-30
A30, F30	AW3D30, FABDEM
E1, E2	endpoints chosen with the truth, or from the coarse map only
Slip model only	the planner without a hazard layer
Gate, weight	planner policies (Section III-D)
Face map (oracle)	the lidar label used as a hazard layer
Mission success	a route that touches no face patch
Vehicle success	mission success with every 5 m step climbable

TABLE II
LIDAR WINDOWS (USGS 3DEP ONE METRE DEM; FLOWN: ACQUISITION PERIOD), FACE PATCHES AND THE SHARE OF CELLS HOLDING A FACE PATCH POST AT 10 AND 30 M.

Site	Lat., lon.	km^2	Relief	Flown	Patches	10 m	30 m
Dumont Dunes, CA	35.65, -116.27	14.4	322 m	2024 to 25	4693	36 %	49 %
White Sands, NM	32.82, -106.29	6.3	20 m	2020	789	15 %	40 %
Kelso Dunes, CA	34.91, -115.72	9.7	240 m	2024 to 25	2184	15 %	38 %
Great Sand Dunes, CO	37.75, -105.54	14.8	246 m	2019 to 20	2065	16 %	32 %
Algodones Dunes, CA	32.92, -115.10	20.0	122 m	2021 to 23	3131	10 %	27 %
Little Sahara, UT	39.70, -112.39	6.4	99 m	2020 to 21	1776	11 %	35 %

window also holds rocky hills, and three windows lie in off road vehicle areas. We chose the windows from 20 m slope images to hold many faces, and had used them in earlier tests of the detector. A post is *steep* when its Horn slope [26] at 1 m is 28° or more, a little below the angle of repose of dry sand [4]; voids never count as steep. A *face patch* is an 8 connected group of at least 10 steep posts (10 m^2). Face patches are steep ground, not verified slip faces. For a coarse cell c with footprint F_c the label is the face map

$$y_c = \begin{cases} 1 & \text{if some } x \in F_c \text{ lies in a face patch,} \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

which a planner needs; its meaning does not change with the cell size, although its prevalence does (Table II).

B. Inputs

Idealised: block means of the lidar at 2 to 60 m (I10 and I30 in the tables), with no product noise and no time gap. *Real:* the USGS 1/3 arc second DEM (R10, about 8×10 m) [27], made from the truth lidar at all six sites (it tests the product chain only); NASADEM (N30, radar of February 2000) [28]; Copernicus GLO-30 (G30, radar of 2011 to 2015) [1]; and, post hoc, AW3D30 (A30) [2] and FABDEM (F30, GLO-30 with forests and buildings removed) [3]. R10, N30 and G30 are scored on their own geographic grids with labels from the lidar posts inside each native cell; A30 and F30, and G30 for planning, are interpolated to the 30 m lidar grid; no shift is applied. *Imagery:* one Sentinel-2 level 2A scene per site [29] (10 m; lidar year, cloud below 5 %, lowest sun) and National Agriculture Imagery Program (NAIP) images [30] (1 m; dates: supplement S5).

C. Hazard layers

The layers (supplement Table S2) are slope, ruggedness [31], plane roughness as in GESTALT [13], an aspect rule, interpolation (the steepest slope after cubic spline interpolation to 1 m), learned super resolution (a dilated network [32] that predicts a 5 m residual), a logistic subgrid detector released with our planner and trained only on synthetic dune profiles, gradient boosted trees [33] in the histogram form of [34] as implemented in scikit-learn [35] on 14 terrain features, a dilated fully convolutional network [36] trained with Adam [37], boosting on terrain plus Sentinel-2 or NAIP features, and, post hoc, a U-Net [38] with early stopping and pixel level networks on the Sentinel-2 and NAIP bands. Every layer trained on lidar is trained leave one site out [39], and a layer that is not a probability is calibrated by logistic regression [40] on the other five sites when a planner needs one.

D. Planner

The planner searches the 8 connected grid of the input. For an edge with along track grade $\tan\alpha$ (uphill positive) and

cross slope $\tan\beta$,

$$d = \sqrt{(\tan\alpha + f_r)^2 + \tan^2\beta}, \qquad i = -i_0 \ln\left(1 - d/\tan\phi\right) \tag{2}$$

give the traction demand and the slip ($i = \infty$ when $d \geq \tan\phi$), with friction angle ϕ, motion resistance coefficient f_r and slip scale i_0: the exponential shear law of Janosi and Hanamoto [41] for the ratio of demand to capacity. An edge is passable when its slip under the nominal parameters is at most $i_{\max}$ and costs $\ell/(1-i)$ for length ℓ. Over 128 parameter draws, a cell is BLOCKED when it is passable (four of eight headings) in at most 5 % of draws, TRAVERSABLE when in at least 95 % and its face probability $p_c < \tau$, and UNKNOWN otherwise. The *posterior vehicle* takes ϕ, f_r and i_0 from a posterior of Eq. (2) fitted by Markov chain Monte Carlo to twelve published slip curves of other wheeled vehicles (test rovers in Mars Yard sand and at Dumont Dunes [5], Scarab and El Dorado [42], Spirit [43]; 5 to 95 % ranges ϕ 30.6 to 43.7°, f_r 0.03 to 0.32, i_0 0.33 to 1.21) and $i_{\max}$ from an assumed range of 0.4 to 0.7 (nominal 36.93°, 0.129, 0.627, 0.5; climbing limit 15.9°). In a leave one study out test on 49 published curves no slip law met our bar (median held out root mean square slip error 0.15; best 0.20), so the vehicle is uncertain; a vehicle with narrower assumed ranges (limit 25.5°) is the sensitivity. Three policies share the graph: *slip only*, *gate* (TRAVERSABLE cells only) and *weight* (no BLOCKED cell, cost times $1 + w\,p_c$ of the cell entered); gate and weight refuse descents of 28° or more. Every layer is run at $\tau \in \{0.02, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 0.9\}$ and $w \in \{1, 2, 5, 10, 20, 50, 100\}$ and reported at the setting with the highest mission success on the other five sites.

E. Scoring and closed loop

One hundred pairs per site, at least 1 km apart, are drawn twice: E1 from 30 m cells whose lidar posts are all below 20° (25° at a site with fewer than 200 such cells; truth used) and E2 from cells whose coarse slope is below 15° (no truth). A route, the polyline through its cell centres sampled every 0.5 m on the lidar, is *unsafe* when a sample falls on a face patch, and a *mission success* when it exists and is not unsafe; *vehicle success* (post hoc) also requires every 5 m step to be passable for the nominal vehicle on the 5 m lidar block mean, with no descent of 28° or more. In the closed loop (20 pairs per site) the rover follows the plan on a 5 m grid that inherits the coarse classes and face probabilities ($w = 5$), with slip costs from the coarse surface interpolated to 5 m. At every step it senses the face cells within $R = 20$ m, blocks them, frees the others and replans when its route is blocked, with up to 300 replans. In the main, post hoc runs a cell is sensed only when the line from an eye 1 m above the rover's cell centre to a point 0.2 m above the cell's centre clears the lidar at 19 equally spaced points; with missed detections each face cell is detectable with probability p_d, drawn once per pair.

F. Statistics and prediction log

Detection is scored by the area under the receiver operating characteristic curve (AUC), with spatial block bootstrap intervals (300 m blocks). The six sites are the units of generalisation: for planning we report the mean of the six per site differences with a t interval (five degrees of freedom). Pairs are not independent (55 % to 82 % of the cells of a slip only route are also used by another route of its site), so we never test at the pair level. An exact sign flip test over six sites reaches at best $p = 1/32$, which any correction for two or more comparisons lifts above 0.05, so we make no significance claims and read each gain by its size, its site level interval and the predictions logged before the runs. The detection results and mission success in Table IV were designed in a prediction log with 43 numeric predictions before they ran (stages 25 to 28), but after earlier results on the same sites were known (31 hits, 12 misses; supplement S1). Vehicle success, the line of sight closed loop, A30, F30, the U-Net, the imagery networks and the face size checks are post hoc and exploratory (stages 29 to 31).

IV. Results

A. Which layer sees faces

Table III gives the mean AUC of every layer for the face map label. On 10 m block means most layers rank face cells well (interpolation 0.97, boosting 0.97, slope 0.90), and R10 behaves like the 10 m block mean (interpolation 0.96). At 30 m every elevation layer loses skill: with default settings interpolation is best (0.82), ahead of boosting (0.81) and the network (0.80), and the detector (0.74) is above only slope, the aspect rule and the NAIP features and network. Super resolution (0.84), the network trained four times longer (0.84), boosting with five more features (0.83) and the U-Net (0.83) do a little better.

Imagery adds a little. Boosting on terrain plus Sentinel-2 features reaches 0.85 at 30 m (+0.04 over terrain alone), with NAIP 0.84; imagery alone is weaker (0.74 and 0.67; pixel level networks 0.78 and 0.62), and at 10 m it adds nothing (0.97). Paired intervals put super resolution above interpolation at six sites and boosting with Sentinel-2 at five, by small margins (supplement Table S4).

The real 30 m products are worse than the 30 m block mean: the best layer reaches 0.72 on GLO-30 and 0.63 on NASADEM on their own grids, and 0.73 on AW3D30 and 0.71 on FABDEM. Interpolation scores 0.712 on AW3D30, 0.678 on FABDEM and 0.696 on GLO-30,; the best shift raises GLO-30 only to 0.702, so misregistration does not explain the loss.

When a 30 m cell needs 100 or 300 steep posts to be positive (fewer count as negative), interpolation scores 0.89 and 0.97, so cells with much steep ground are ranked well; other face rules change little (supplement S6).

B. How fine a map must be

Fig. 2a shows how detection falls with grid spacing: interpolation scores 1.00 at 2 m, 0.97 at 10 m, 0.82 at 30 m and 0.69 at 60 m. Fig. 2b relates the fall to patch size. At every spacing boosting finds nearly all patches longer than four grid cells (96 % of patches 4 to 8 cells long at 30 m), but at 30

TABLE III

DETECTION OF THE FACE MAP LABEL OF EQ. (1): MEAN AUC OVER THE SIX SITES. I: LIDAR BLOCK MEANS; I10→30: THE LARGEST OF THE NINE 10 M SCORES IN EACH 30 M CELL; R10: USGS 1/3 ARC SECOND DEM, MADE BY USGS FROM THE SAME LIDAR AT ALL SIX SITES; N30: NASADEM; G30: GLO-30; R10, N30 AND G30 ON THEIR OWN GRIDS, WHERE THE NETWORKS AND SUPER RESOLUTION WERE NOT RUN (ON GLO-30 INTERPOLATED TO THE 30 M LIDAR GRID THEY REACH 0.72 AND 0.70); A30: AW3D30 AND F30: FABDEM, INTERPOLATED TO THE 30 M LIDAR GRID. RUGGEDNESS AND ROUGHNESS (I30 0.76 AND 0.77): SUPPLEMENT TABLE S5. *TRAINED ON THE LIDAR OF THE OTHER FIVE SITES. ‡POST HOC. BEST PER COLUMN IN BOLD (TIES AT TWO DECIMALS ALL BOLD); · NOT RUN.

Layer	I10	I30	I10→30	R10	N30	G30	A30‡	F30‡
Slope	0.90	0.69	·	0.90	0.62	0.65	0.66	0.64
Aspect rule	0.83	0.61	·	·	·	·	·	·
Interpolation	**0.97**	0.82	**0.96**	**0.96**	0.62	0.70	0.71	0.68
Super resolution*	·	0.84	·	·	·	·	·	·
Detector	0.95	0.74	0.95	0.95	0.56	0.65	0.65	0.64
Boosting*	**0.97**	0.81	**0.96**	0.95	**0.63**	**0.72**	**0.73**	**0.71**
Boosting+*	**0.97**	0.83	·	·	·	·	·	·
Network*	0.96	0.80	0.95	·	·	·	·	·
Network, 6000 steps*	·	0.84	·	·	·	·	·	·
Sentinel-2 only*	0.68	0.74	·	·	·	·	·	·
Boosting + Sentinel-2*	**0.97**	**0.85**	·	·	·	·	·	·
NAIP only*	0.62	0.67	·	·	·	·	·	·
Boosting + NAIP*	**0.97**	0.84	·	·	·	·	·	·
Sentinel-2 network*‡	·	0.78	·	·	·	·	·	·
NAIP network*‡	·	0.62	·	·	·	·	·	·
U-Net*‡	·	0.83	·	·	·	·	·	·

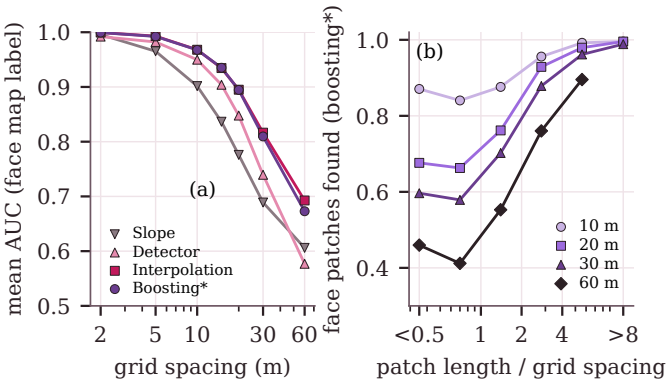


Fig. 2. Resolution and scale (lidar block means, face map label). (a) Mean AUC over six sites against the grid spacing. (b) Share of face patches found by boosting (a cell holding the patch is flagged at the threshold that flags 10 % of face free cells) against patch length over grid spacing, in the bins below 0.5, 0.5 to 1, 1 to 2, 2 to 4, 4 to 8 and above 8 (plotted at 0.35, 0.7, 1.4, 2.8, 5.6 and 11.3), for four spacings; bins with fewer than 50 patches are omitted. Patch length: major axis of the ellipse with the same second moments. *Trained leave one site out.

and 60 m far fewer patches shorter than half a cell (60 % at 30 m, 46 % at 60 m) (longer patches also touch more cells). By count most patches are short: 75 % are shorter than one 30 m cell, but they hold only 7 % of the face posts.

C. Planning

Table IV gives mission and vehicle success. The gate routes at most 34 % of pairs at 10 m and 42 % at 30 m: the slip model is uncertain on most dune slopes, so even gentle E1 endpoints are often not TRAVERSABLE on the coarse map (76 % of the refusals of boosting at 30 m and $\tau = 0.2$ are at an endpoint).

TABLE IV

MISSION SUCCESS / VEHICLE SUCCESS (%; VEHICLE SUCCESS POST HOC) AT THE OPERATING POINT CHOSEN ON THE OTHER FIVE SITES, 100 PAIRS PER SITE, POSTERIOR VEHICLE. E1: ENDPOINTS CHOSEN WITH THE TRUTH; E2: FROM THE COARSE MAP ONLY. SITE LEVEL INTERVALS: SUPPLEMENT S6. *TRAINED LEAVE ONE SITE OUT.

	I10		I30		G30
Layer	E1	E2	E1	E2	E1
Slip model only	17/9	9/4	8/6	3/1	6/4
Slope	35/20	28/15	6/4	3/1	6/4
Interpolation	64/47	53/34	13/7	5/2	7/4
Detector	50/30	33/18	7/4	4/2	5/4
Boosting*	59/39	47/26	9/6	4/2	7/5
Network*	51/33	40/21	6/4	3/2	7/4
Boosting + Sentinel-2*	56/37	46/23	13/9	7/4	·
Face map (oracle)	84/25	80/18	81/17	68/11	80/15

The weight policy routes almost every pair and is chosen for every layer except slope and the detector on GLO-30, mostly at the largest weight on 10 m maps; with weights up to 1000 (post hoc) success rises by up to +12 points at 10 m and +1 at 30 m.

On 10 m maps every layer helps: mission success rises from 17 % with the slip model alone to 64 % with interpolation (+47 points, site level interval +22 to +73), and vehicle success from 9 % to 47 %. On 30 m maps the gain is small: interpolation adds +5 mission points (interval +1 to +9) and boosting with Sentinel-2 +5 (0 to +10) to 8 %, but no computable layer adds more than 2.5 points of vehicle success, and slope, the detector and the network fall below the slip model alone: they flag much of the dune, so routes grow longer and still meet faces. Every layer trades length for exposure: the slip model alone drives 1.19 times the straight distance with 33 m on faces per km, interpolation 1.37 times with 11 m and the oracle 1.53 times with 2 m. One contact fails a route, so cutting exposure by two thirds still leaves most routes failing. On GLO-30 no layer adds more than 1.7 points, on AW3D30 interpolation adds 2.0 and on FABDEM 0.2 (supplement Table S13). The sensitivity vehicle and pair distance change little (Tables S14 and S10). With E2 endpoints the slip model alone reaches 3 % at 30 m and the gain of interpolation is +3 (interval 0 to +6).

The face map oracle reaches 81 % mission success on the 30 m grid but only 17 % vehicle success: 21 % of its successes are drivable, because its routes go round faces over slopes the coarse map shows as gentle. Registered degraded face maps at $w = 5$ (supplement Table S7) show how exact face information must be: the oracle gives 55 % at this weight, displaced by 10 m 43 % and by 30 m 11 %, and maps from simulated 2 m and 5 m surveys only 23 % and 8 %, because the 28 degree rule misses faces on a coarser grid.

Face size. Scored against faces of at least 60 m only, 30 m interpolation adds +8 points over the slip model alone (29 % to 37 %), against +42 on 10 m maps; against patches of at least 300 posts it adds +13 (30 % to 43 %; supplement Table S9). The 30 m gap is smaller for large faces but remains. *Stronger learned layers.* The U-Net adds +2 points at 30 m (9 %; interval −1 to +4), and the NAIP network −1.

TABLE V
EXPLORATORY CLOSED LOOP, SENSING WITHIN 20 M (POST HOC EXCEPT THE LAST COLUMN), 20 PAIRS PER SITE, $w = 5$: SUCCESS (%; GOAL REACHED, NO FACE TOUCHED). LINE OF SIGHT: EYE 1 M ABOVE THE GROUND; $p_d = 0.8$: EACH FACE CELL DETECTABLE WITH PROBABILITY 0.8. IDEAL: THE REGISTERED SENSOR THAT SEES EVERY FACE. INTERVALS AND SENSOR VARIANTS: SUPPLEMENT S6. *TRAINED LEAVE ONE SITE OUT.

	Line of sight				Ideal
	I30			I10	I30
Prior layer	E1	E2	E1, p_d 0.8	E1	E1
Slip model only	30	29	12	41	80
Detector	34	35	·	·	85
Interpolation	48	36	·	63	82
Boosting*	42	38	21	·	84
Boosting + Sentinel-2*	48	36	·	·	·
Super resolution*	48	36	·	·	·
Network, 6000 steps*	38	34	·	·	·
Face map (oracle)	74	64	68	82	85

D. Onboard sensing

Table V adds onboard sensing within 20 m (exploratory). A slip face lies beyond a brink, out of sight of an approaching rover (Fig. 1d), so a 5 m cell is sensed only when in line of sight. With the slip model alone as 30 m prior the rover succeeds on 30 % of the E1 pairs (65 % touch a face); the computable priors give 34 % (detector) to 48 % (interpolation, gain +18 points, site level interval +2 to +34) and the oracle 74 %. With E2 endpoints no computable prior has a site level interval above zero (best boosting, 38 % against 29 %). With the weight chosen on the other five sites instead of $w = 5$, interpolation gives 53 % and the oracle 81 %. The sensor matters (supplement S6): with the slip model prior an eye at 0.5 or 1.5 m gives 28 % and 42 %, and a detection probability of 0.8 drawn once per pair 12 %. On 10 m maps interpolation and the oracle give 63 % and 82 % at $w = 5$, below the open loop (64 % and 84 %), but 78 % and 85 % with the weight chosen on the other sites, so that gap came from the fixed weight.

V. DISCUSSION AND LIMITATIONS

What the results mean. A 30 m elevation model keeps enough of the dune shape to rank face cells, but not enough to plan around faces: one missed face fails a route of a kilometre, so recall, not ranking, decides planning, and the gap to 10 m maps remains even for long faces. A 30 m map also hides the steep climbs between faces, so even the face map oracle drives few routes the vehicle can climb. In the exploratory closed loop a coarse prior helps with E1 endpoints but not clearly with E2 endpoints, and the result moves with the sensor. An exact face map more than doubles success, but no computable layer approaches it, and survey face maps lose most of its value unless the face rule is recalibrated.

Limitations. The truth is steep ground: rock, ruts and vegetated mounds also exceed 28 degrees, three windows are off road vehicle areas, and the audit of 600 patches is not finished, so label precision is unknown. We chose the windows from slope images and had used them before, so earlier design choices may fit these sites; the prediction log is in our own repository, and much of the analysis is post hoc. The vehicle is a posterior fitted to slip curves of other vehicles and no slip law passed our transfer test, so absolute outcomes are uncertain. No rover was driven, the sensor model is simple, and the 10 m results use lidar block means. Six sites are few: the site level intervals have five degrees of freedom, one unusual site such as Dumont can move a mean by several points, and the exact percentages are bounded by this sample. All sites are in the United States; on Arabian fields a megadune face 100 m high spans about five 30 m cells, but smaller flank faces would not.

VI. CONCLUSION

A 30 m elevation model ranks slip faces better than chance, but a planner using it does not keep a small rover off them. On six lidar dune fields no hazard layer computable from such a map, including imagery fusion, super resolution and a U-Net, added more than 5.3 points of mission success at its held out operating point, and real 30 m products did worse, while interpolation on a 10 m map raised it from 17 % to 64 %; the gap remains even for long faces and when the vehicle must also climb every step. In exploratory runs with line of sight sensing a 30 m prior helped by +4 to +18 points. Next are the label audit, Arabian fields with a drone survey, and a rover with measured slip on the same sand.

DATA AND CODE

USGS 3DEP lidar and DEMs, NASADEM and NAIP are in the public domain; Sentinel-2 data are provided under the Copernicus free licence; AW3D30 is © JAXA; Copernicus GLO-30 is used under its licence, and FABDEM (University of Bristol) under CC BY-NC-SA 4.0 for non commercial research (both produced using Copernicus WorldDEM-30 © DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018, provided under COPERNICUS by the European Union and ESA; all rights reserved). Code, results, the prediction log (stages 25 to 31) and the scripts that regenerate every number and figure are at https://github.com/hudasol/dune-rover.

REFERENCES

[1] European Space Agency and Airbus, "Copernicus DEM," Dataset, European Space Agency, 2022. [Online]. Available: https://doi.org/10.5270/ESA-c5d3d65

[2] T. Tadono, H. Ishida, F. Oda, S. Naito, K. Minakawa, and H. Iwamoto, "Precise global DEM generation by ALOS PRISM," *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. II-4, pp. 71–76, 2014.

[3] L. Hawker, P. Uhe, L. Paulo, J. Sosa, J. Savage, C. Sampson, and J. Neal, "A 30 m global map of elevation with forests and buildings removed," *Environmental Research Letters*, vol. 17, no. 2, p. 024016, 2022.

[4] K. Pye and H. Tsoar, *Aeolian Sand and Sand Dunes*, 2nd ed. Berlin: Springer, 2009.

[5] M. Heverly, J. Matthews, J. Lin, D. Fuller, M. Maimone, J. Biesiadecki, and J. Leichty, "Traverse performance characterization for the Mars Science Laboratory rover," *Journal of Field Robotics*, vol. 30, no. 6, pp. 835–846, 2013.

[6] R. E. Arvidson, K. D. Iagnemma, M. Maimone, A. A. Fraeman, F. Zhou, M. C. Heverly, P. Bellutta, D. Rubin, N. T. Stein, J. P. Grotzinger, and A. R. Vasavada, "Mars Science Laboratory Curiosity rover megaripple crossings up to sol 710 in Gale crater," *Journal of Field Robotics*, vol. 34, no. 3, pp. 495–518, 2017.

[7] W. Zhang and D. R. Montgomery, "Digital elevation model grid size, landscape representation, and hydrologic simulations," *Water Resources Research*, vol. 30, no. 4, pp. 1019–1028, 1994.

[8] N. Vandapel, R. R. Donamukkala, and M. Hebert, "Unmanned ground vehicle navigation using aerial ladar data," *The International Journal of Robotics Research*, vol. 25, no. 1, pp. 31–51, 2006.

[9] J. Delmerico, E. Mueggler, J. Nitsch, and D. Scaramuzza, "Active autonomous aerial exploration for ground robot path planning," *IEEE Robotics and Automation Letters*, vol. 2, no. 2, pp. 664–671, 2017.

[10] J.-M. Fortin, O. Gamache, W. Fecteau, E. Daum, W. Larrivée-Hardy, F. Pomerleau, and P. Giguère, "UAV-assisted self-supervised terrain awareness for off-road navigation," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 1–7.

[11] M. Ono, B. Rothrock, E. Almeida, A. Ansar, R. Otero, A. Huertas, and M. Heverly, "Data-driven surface traversability analysis for Mars 2020 landing site selection," in *Proc. IEEE Aerospace Conference*, 2016, pp. 1–12.

[12] P. Papadakis, "Terrain traversability analysis methods for unmanned ground vehicles: A survey," *Engineering Applications of Artificial Intelligence*, vol. 26, no. 4, pp. 1373–1385, 2013.

[13] S. B. Goldberg, M. W. Maimone, and L. Matthies, "Stereo vision and rover navigation software for planetary exploration," in *Proc. IEEE Aerospace Conference*, vol. 5, Big Sky, MT, USA, 2002, pp. 2025–2036.

[14] G. Ishigami, A. Miwa, K. Nagatani, and K. Yoshida, "Terramechanics-based model for steering maneuver of planetary exploration rovers on loose soil," *Journal of Field Robotics*, vol. 24, no. 3, pp. 233–250, 2007.

[15] S. Shrivastava, A. Karsai, Y. O. Aydin, R. Pettinger, W. Bluethmann, R. O. Ambrose, and D. I. Goldman, "Material remodeling and unconventional gaits facilitate locomotion of a robophysical rover over granular terrain," *Science Robotics*, vol. 5, no. 42, p. eaba3499, 2020.

[16] A. Angelova, L. Matthies, D. Helmick, and P. Perona, "Learning and prediction of slip from visual information," *Journal of Field Robotics*, vol. 24, no. 3, pp. 205–231, 2007.

[17] S. Thrun, M. Montemerlo, H. Dahlkamp, D. Stavens, A. Aron, J. Diebel, P. Fong, J. Gale, M. Halpenny, G. Hoffmann, K. Lau, C. Oakley, M. Palatucci, V. Pratt, P. Stang, S. Strohband, C. Dupont, L.-E. Jendrossek, C. Koelen, C. Markey, C. Rummel, J. van Niekerk, E. Jensen, P. Alessandrini, G. Bradski, B. Davies, S. Ettinger, A. Kaehler, A. Nefian, and P. Mahoney, "Stanley: The robot that won the DARPA Grand Challenge," *Journal of Field Robotics*, vol. 23, no. 9, pp. 661–692, 2006.

[18] H. Inotsume, T. Kubota, and D. Wettergreen, "Robust path planning for slope traversing under uncertainty in slip prediction," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 3390–3397, 2020.

[19] C. H. Hugenholtz, N. Levin, T. E. Barchyn, and M. C. Baddock, "Remote sensing and spatial analysis of aeolian sand dunes: A review and outlook," *Earth-Science Reviews*, vol. 111, no. 3–4, pp. 319–334, 2012.

[20] J. Daynac, P. Bessin, S. Pochat, R. Mourgues, and S. Shumack, "A new workflow for mapping dune features (outline, crestline and defects) combining deep learning and skeletonization from DEM-derived data," *Geomorphology*, vol. 463, p. 109369, 2024.

[21] Z. Chen, X. Wang, Z. Xu, and W. Hou, "Convolutional neural network based DEM super resolution," in *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XLI-B3, 2016, pp. 247–250.

[22] P. L. Guth and T. M. Geoffroy, "LiDAR point cloud and ICESat-2 evaluation of 1 second global digital elevation models: Copernicus wins," *Transactions in GIS*, vol. 25, no. 5, pp. 2245–2261, 2021.

[23] M. Martone, B. Bräutigam, P. Rizzoli, N. Yague-Martinez, and G. Krieger, "Enhancing interferometric SAR performance over sandy areas: Experience from the TanDEM-X mission," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 9, no. 3, pp. 1036–1046, 2016.

[24] M. Wang, H. Li, Y. Liu, and H. Li, "Multi-source DEM vertical accuracy evaluation of Taklimakan desert hinterland based on ICESat-2 ATL08 and UAV data," *Remote Sensing*, vol. 17, no. 11, p. 1807, 2025.

[25] L. J. Sugarbaker, E. W. Constance, H. K. Heidemann, A. L. Jason, V. Lukas, D. L. Saghy, and J. M. Stoker, "The 3D Elevation Program initiative: A call for action," U.S. Geological Survey, Circular 1399, 2014.

[26] B. K. P. Horn, "Hill shading and the reflectance map," *Proceedings of the IEEE*, vol. 69, no. 1, pp. 14–47, 1981.

[27] U.S. Geological Survey, "1/3 arc second digital elevation models, 3D Elevation Program," Dataset, The National Map, 2026, accessed October 2026. [Online]. Available: https://apps.nationalmap.gov/downloader/

[28] NASA JPL, "NASADEM merged DEM global 1 arc second V001," Dataset, NASA EOSDIS Land Processes Distributed Active Archive Center, 2020. [Online]. Available: https://doi.org/10.5067/MEaSUREs/NASADEM/NASADEM_HGT.001

[29] M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, B. Hoersch, C. Isola, P. Laberinti, P. Martimort, A. Meygret, F. Spoto, O. Sy, F. Marchese, and P. Bargellini, "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," *Remote Sensing of Environment*, vol. 120, pp. 25–36, 2012.

[30] U.S. Department of Agriculture, Farm Service Agency, "National agriculture imagery program (NAIP)," Dataset, read through the Microsoft Planetary Computer, 2026, accessed October 2026. [Online]. Available: https://planetarycomputer.microsoft.com/dataset/naip

[31] S. J. Riley, S. D. DeGloria, and R. Elliot, "A terrain ruggedness index that quantifies topographic heterogeneity," *Intermountain Journal of Sciences*, vol. 5, no. 1-4, pp. 23–27, 1999. [Online]. Available: https://arc.lib.montana.edu/ojs/index.php/IJS/article/view/2914

[32] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," in *International Conference on Learning Representations (ICLR)*, 2016. [Online]. Available: https://arxiv.org/abs/1511.07122

[33] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.

[34] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

[35] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, no. 85, pp. 2825–2830, 2011. [Online]. Available: https://jmlr.org/papers/v12/pedregosa11a.html

[36] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 3431–3440.

[37] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations (ICLR)*, 2015. [Online]. Available: https://arxiv.org/abs/1412.6980

[38] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, ser. Lecture Notes in Computer Science, vol. 9351, 2015, pp. 234–241.

[39] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, and C. F. Dormann, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017.

[40] J. C. Platt, "Probabilities for SV machines," in *Advances in Large-Margin Classifiers*, A. J. Smola, P. Bartlett, B. Schölkopf, and D. Schuurmans, Eds. Cambridge, MA: The MIT Press, 2000, pp. 61–74.

[41] Z. Janosi and B. Hanamoto, "The analytical determination of drawbar pull as a function of slip for tracked vehicles in deformable soils," in *Proc. 1st International Conference on the Mechanics of Soil Vehicle Systems*. Turin, Italy: Edizioni Minerva Tecnica, 1961.

[42] H. Inotsume, "Analysis of angle of attack for efficient slope ascent by rovers," Master's thesis, Robotics Institute, Carnegie Mellon University, Aug. 2016, Tech. Rep. CMU-RI-TR-15-22. [Online]. Available: https://publications.ri.cmu.edu/storage/publications/pub_files/2016/8/MSThesis_Hiroaki.pdf

[43] R. Li, B. Wu, K. Di, A. Angelova, R. E. Arvidson, I.-C. Lee, M. Maimone, L. H. Matthies, L. Richter, R. Sullivan, M. H. Sims, R. Greenberger, and S. W. Squyres, "Characterization of traverse slippage experienced by Spirit rover on Husband Hill at Gusev crater," *Journal of Geophysical Research: Planets*, vol. 113, no. E12, 2008.